

COMPUTER PROGRAMMING

LAB REPORT 11

Title: String Handling in C++

1. OBJECTIVE

- To understand the concept of strings in C++
- To learn string declaration and initialization
- To practice string concatenation
- To calculate and display string length

2. THEORY

Strings in C++ are used to store text data such as names, messages, and sentences. The string data type is available through the <string> library. Different operations can be performed on strings including concatenation, displaying text, and finding string length.

3. SOFTWARE USED

- DEV-C++ IDE
- GNU C++ Compiler

4. PROGRAMS

Program 1 — Full Name Concatenation

Code

```
#include <iostream>    // header file is included for input and output
#include <string>       // string library is included
using namespace std;  // standard namespace is used
int main()
{
    string firstName = "Danyal";    // first name string is declared
    string lastName = "Hassan";    // last name string is declared
    string fullName = firstName + " " + lastName;
    cout << "Full Name: " << fullName << endl; // full name is displayed on screen
    return 0;
}
```

Output

Full Name: Danyal Hassan

Program 2 — String Length Calculation

Code

```
#include <iostream> // header file is included for input and output
#include <string>    // string library is included
using namespace std; // standard namespace is used
int main()          // main function starts program execution
{
    string message = "Welcome to C++ Programming";
    // string variable is declared and initialized
    cout << "Message: " << message << endl; // message is displayed on
    screen
    cout << "Length of string = " << message.length() << endl;
    // length of string is displayed
    return 0;
}
```

Output

Message: Welcome to C++ Programming
Length of string = 26

5. CONCLUSION

In this lab, we learned how to use strings in C++ for storing and displaying text. Different string operations such as concatenation and finding string length were performed successfully. This lab improved understanding of text handling and basic string manipulation in C++.